Air Liquide (AI.PA) — Valuation Report

Valuation date 30 October 2026 · Route: published_damodaran · Academic analysis for educational purposes; not personal investment advice.

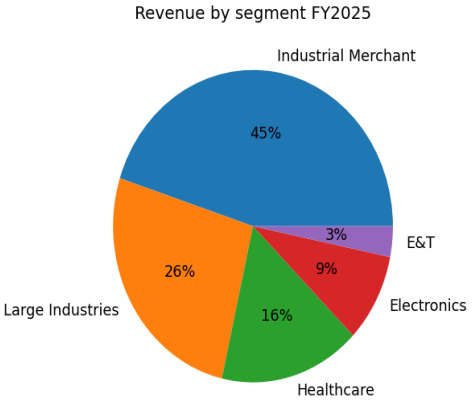
1. One-page summary

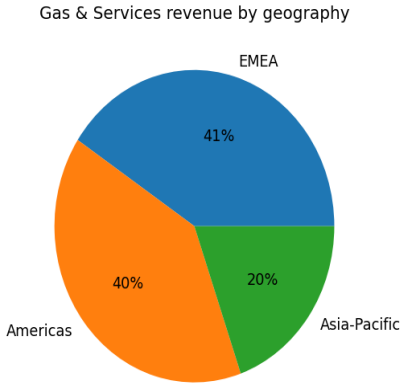
Recommendation: SELL. Fair value range **€80–170**, central estimate €110; 12-month recommended price **€110** vs market €180.9 → –**39%**. Key drivers: (1) WACC 7.70% vs the market's implied ~5% cost of capital; (2) OIR margin held at a flat 20.7% — the market prices durable expansion; (3) mid-single-digit organic growth, not the ~8% priced into peer multiples.

In plain English: Air Liquide is a high-quality, defensive business — that is exactly the problem. The market already pays a premium multiple for stability. Our three independent methods all land between roughly €80 and €210, with the centre of gravity well below the current price. Quality is real; the price already includes it.

2. Company in 5 minutes

Air Liquide supplies industrial and medical gases (oxygen, nitrogen, hydrogen, specialty gases) under long-term contracts and on-site arrangements, plus engineering & technology and home healthcare. Five segments: Industrial Merchant (€12.1bn), Large Industries (€7.1bn), Healthcare (€4.4bn), Electronics (€2.5bn), E&T; (€0.9bn). Gas & Services revenue splits EMEA €10.6bn / Americas €10.4bn / Asia-Pacific €5.1bn.





3. Industry and competitors

Industrial gases is a global oligopoly (Linde, Air Liquide, Air Products, Nippon Sanso) with high barriers: on-site pipelines create local monopolies, energy pass-through clauses stabilise margins, and the business is capital-intensive but defensive through cycles. Growth tracks industrial production plus structural demand (semiconductors, healthcare, hydrogen).

4. Financial history

FY2022–25 revenue grew from €29.9bn to €26.9bn reported (lower energy pass-through masks +4-5% organic growth); OIR margin expanded to 20.7%; EPS €6.10 (restated after the June-2026 1:10 free-share attribution). Anomaly report: cash +107% YoY (2026 bond prefunding), working-capital sign flips on a near-zero base, lumpy M&A; and paused buybacks — all kept as reported; none requires normalisation. Interim June-2026 net debt stepped to €15.2bn (DIG Airgas acquisition + prefunding) vs €8.4bn at FY25 — this debt-path instability is why the DCF uses FCFF.

5. Forecast assumptions

• Sales +4%/yr for 5 years — in line with guidance and the defensive demand mix.

• OIR margin 20.7% flat — no further expansion assumed (market likely prices more).

• NCE/NOPAT 38% — recent net capex intensity sustained.

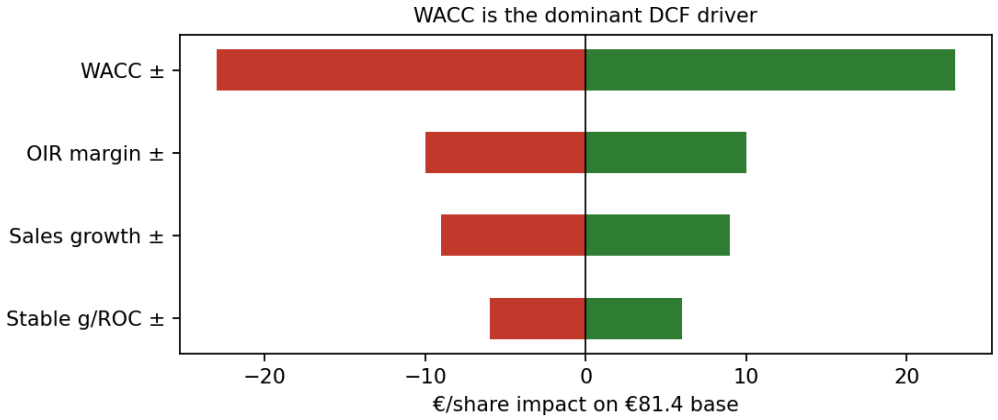
• Selected NI growth ~5% — fundamental (ROE×(1–payout) ≈ 4.9–5.3%) vs analysts ~5–6%.

• Stable phase: g = 3.0% (≤ Rf), stable ROC 10.7% — gas economics are good but not monopolistic.

6. DCF valuation (FCFF)

Rf 3.38% (Bund 10y); mature ERP 4.20%; weighted country-risk premium 0.79% (Americas 0 / EMEA 0.92 / ME&A; 2.99 / Asia 1.03 proxies); bottom-up unlevered β 0.891 from 5 segments, relevered βL 0.970 at market D/E 11.85% (observed 5Y regression β 0.61 disclosed as cross-check, beta-scaled Ke 8.22%). Ke 8.24%; Kd 4.16% (actual A2/A spread 0.78%; synthetic AAA cross-check); **WACC 7.70%**.

Result: **€81.4/share** (PV of terminal value 85.7% — flagged). LOW 58.1 / HIGH 113.4. Drivers: WACC ±€23, OIR margin ±€10, sales growth ±€9.



WACC vs terminal growth (€/share):

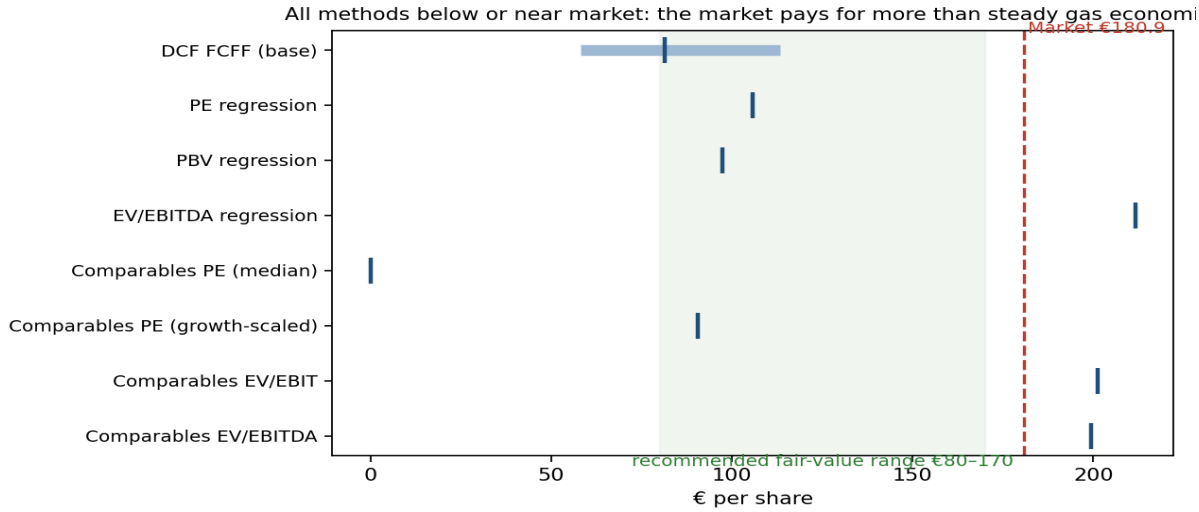
WACC \ g	2.0%	3.0%	4.0%
7.2%	91	103	120
7.7%	72	81	93
8.2%	58	65	73

7. Relative valuation

7a. Comparable companies — 10 peers (all Tier 1–2, 7 Europe-listed): Linde, Air Products, Nippon Sanso (world expansion), Givaudan, BASF, Alfa Laval, EMS-Chemie, Symrise, Yara, DSM-Firmenich. Documented exclusions: SOL (size band), Novonesis and Syensqo (restructuring inside the comparison period).

Primary multiple PE (gases/machinery mapping): median 30.9 (n=8) → **€171/sh raw**; growth-scaled PE 16.4 → €91. Cross-checks EV/EBIT 27.0 → €201, EV/EBITDA 18.7 → €199. Quartile range €80–224.

7b. Published Damodaran regressions (EUROPE, Jan-2026 vintage): PE = 9.72 + 2.99β + 49.82·g_eps + 7.98·payout → 19.14 → **€105.8**; PBV → 2.36 → **€97.3**; EV/EBITDA → 19.73 → **€211.6**. g_eps/g_rev are approved fundamental substitutions; SEy/residuals n/a_published_route; vintage ~10 months old (limitation).



8. Scenarios and sensitivities

DCF: bear €58.1 (WACC 8.2%, margin –100bp, g 2%), base €81.4, bull €113.4 (WACC 7.2%, margin +100bp, g 4%). Comparables: peer-quartile span €80–224; ex-world-expansion subset (7 European peers) PE median ~28 → ~€155. Regression: PE/PBV cluster €97–106 vs EV/EBITDA €212 — the EV/EBITDA equation's g_rev term carries the optimism. Most likely scenario: base — Air Liquide's history supports steady, not accelerating, economics.

9. Reconciliation

Method	Value €/sh	Core assumption that differs
DCF FCFF	81.4	7.70% WACC, flat 20.7% margin, 3% terminal growth — conservative cost of capital
Comparables PE (raw)	170.8	Peers deserve 30.9× earnings — mixes higher-margin franchises (Givaudan, EMS, Symrise)
Comparables PE (scaled)	90.7	PE scaled to AI's own ~5% fundamental growth
Regression PE / PBV	105.8 / 97.3	Market-wide EUROPE equations: price AI as an average European firm
Regression EV/EBITDA	211.6	g_rev term + peer EBITDA margins above AI's 28%
Market	180.9	~6–8% durable growth at a ~5% implied cost of equity

The DCF-multiples gap is mostly the discount rate: closing it requires Ke ≈ 5% (implying β≈0.4 with this CRP — below the observed 0.61) or margins ~23% sustained. Peers trade on quality-premium multiples the regression's market-average fit does not reproduce; the two EV-based numbers inherit both that premium and the definitional EBITDA spread.

10. Market expectations and current conditions

At €180.9 and 32.6× earnings the market prices Air Liquide as a durable-compounder franchise: ~6–8% EPS growth indefinitely, expanding margins, and a cost of equity near 5%. Defensive positioning amid high European energy costs supports the premium today. That premium is what our model calls into question — not the business.

11. FINAL CONCLUSION

Fair value: low €80 / base €110 / high €170. Recommended 12-month price **€110**. Synthesis: method-suitability weighting favouring DCF and the growth-adjusted PE (accounting-basis consistency) over raw peer medians that import foreign margin structures — stated ex ante, not tuned to the market price.

Mispricing: market €180.9 vs fair value €110 → overvalued by ~€71, –**39.2%**.

Limitations: FY2021 column aggregator-only; external implied-Ke source unavailable; NCI and pensions at book proxies; regression vintage Jan-2026; peer multiples single-source (stockanalysis); Spot_check human review pending.

Recommendation: SELL. The market is most likely mispricing **margin durability**: the annual report frames DIG Airgas synergies and efficiency programs, and consensus extrapolates them; our flat-margin, market-WACC model shows even a well-run Air Liquide is worth far less than the price asks.



Analyst comparison: stockanalysis consensus (20 analysts) = “Strong Buy”, mean target €200.29 — single aggregator source; Morningstar/LSEG/Bloomberg targets were not accessible. Noted, not weighted.

12. Risks — what would change the view

• DIG Airgas integration delivering faster synergy capture (→ closer to bull €113+).

• A lower measured cost of equity (β evidence vs 0.61 observed) would lift the DCF toward €100+.

• Energy-cost deflation + merchant pricing could validate the peer multiple premium.

• Conversely: European industrial slowdown hits Industrial Merchant volumes (largest segment).

13. Glossary

EV enterprise value · WACC weighted average cost of capital · EBITDA/EBIT earnings before interest, tax, D&A; / before interest and tax · FCFF free cash flow to the firm · OIR operating income recurring (Air Liquide's adjusted operating profit) · CRP country risk premium · terminal value = value of cash flows beyond the forecast horizon · peer tier = closeness of business similarity · published regression = Damodaran's pre-estimated market-wide multiple equations.

14. Appendices

A. DCF schedules: leverage history & DIG Airgas debt step; reported-vs-adjusted bridge (OIR 5,581.6 vs IFRS op. income 5,278.7 vs aggregator 5,201); effective tax 25.8%; NCE/NWC ratios; reinvestment-growth identity; segment-beta and CRP tables; WACC bridge; post-attribution share count 636.47m; conventions/conflicts table. See dcf_decisions.md / dcf_model.xlsx.

B. Relative evidence: comparables.csv (30 candidates, scores, exclusions), multiples_summary.csv, comparables_audit.xlsx (spot-check diff ≤0.2%), regression_valuation.xlsx, valuation_multiples.csv — all under relative/.

C. Anomaly report summary (14 flags, all kept); data gaps; source_log.csv 204 dated rows.

Disclaimer: academic analysis for educational purposes; not personal investment advice.